REST Countries Data Profiling Report

Milestone 1 | Raw data profiling | 254 country and territory records

Scope

This report summarizes the supplied profiling results for a tabular dataset obtained from the REST Countries API. It documents field representations, completeness, descriptive statistics, outliers, regional coverage, duplicates, and validity checks. The report is limited to profiling; it does not prescribe or claim that cleaning has been performed.

1. Data overview

Column	Data type / representation
CountryName	Object (text)
Region	Object (text)
Area(km2)	Float; square kilometers
Languages	Text; comma-separated language names
CurrencyCodes	Text; comma-separated currency codes
borders	Text; comma-separated land-neighbor codes
Population	Integer; population count

2. Completeness checks

Empty strings were checked with df[column] == "" for the three text-list fields. Null values for the other four columns were checked with isna().sum(). These are separate tests and are reported separately.

Column	Empty strings	Share of 254
Languages	1	0.39%
CurrencyCodes	3	1.18%
borders	86	33.86%

Null-value check: CountryName, Region, Area(km2), and Population each have 0 null values (0.00% of 254 records).

Interpretation: An empty borders string can legitimately describe a country or territory with no land neighbors; it is not necessarily a missing-data error. The empty-string checks above do not establish whether other fields contain nulls, nor do the four null checks establish whether their fields contain empty strings.

3. Descriptive statistics

Statistic	Area (km²)	Population
Count	254	254
Mean	591,731.2	31,779,170
Standard deviation	1,895,959	131,409,200
Minimum	0.44	0
Q1 (25%)	1,439.75	222,007.8
Median (50%)	62,979	4,920,816
Q3 (75%)	353,423.8	20,540,570
Maximum	17,098,240	1,429,404,000

Values are reproduced using the precision shown in the supplied pandas describe() output.

Both distributions show substantial variation: the means greatly exceed the medians, consistent with positive skew and a relatively small number of very large countries or populations. The observed minimum population is zero.

4. Outlier screening

The 1.5 × interquartile range (IQR) rule was applied separately to each numeric column. A flag identifies a statistically extreme value; it does not, by itself, mean the record is incorrect.

Numeric field	Flagged records	Share of 254
Area(km2)	35	13.78%
Population	30	11.81%

The two flag counts cannot be added to obtain a unique record total, because a record may be flagged in both columns.

5. Records by API region

Region	Records	Share of 254
Africa	60	23.62%
Americas	56	22.05%
Europe	54	21.26%
Asia	52	20.47%
Oceania	27	10.63%
Antarctic	5	1.97%
Total	254	100.00%

The regional categories are those returned by the API. Africa has the highest record count (60), while Antarctic has the lowest (5). Percentages are computed as region count ÷ 254 × 100 and rounded to two decimal places.

6. Duplicate and validity checks

Check	Finding
Duplicate full rows	0
Area ≤ 0	0
Population < 0	0
Population = 0	2 records (0.79%)

The two zero-population records are Bouvet Island and Heard Island and McDonald Islands. A population of zero is not a negative value and need not indicate a data error.

7. Main findings and interpretation

• Coverage and numeric completeness are strong within the checked fields: 254 observations have reported area and population values, with no nulls in the four columns tested.

• The most frequent empty-string field is borders (86 records; 33.86%), but absence of land neighbors may be meaningful rather than defective.

• Area and population are highly dispersed and right-skewed; the IQR rule flags 35 area values and 30 population values for review, not automatic removal.

• No duplicate full rows, nonpositive areas, or negative populations were reported. Two records have zero population and should be interpreted in their geographic context.

Calculation note: All percentages in this report are derived from the supplied counts using a denominator of 254 and are rounded to two decimal places. All other numerical values are reproduced from the supplied profiling results.